Insta-V2 Recommendation Engine

Mathematical Specification and Algorithmic Logic Sheet

Production Reference for Python / Django ML Microservice

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Abstract

This specification documents the complete mathematical foundations, feature-space representations, scoring equations, and ranking logic implemented in the Insta-V2 Machine Learning Recommendation Microservice (`recommendations/engine.py`). The microservice exposes three independent personalization pipelines: Recommended Users (`user-graph-interest-v1`), Recommended Posts (`hybrid-tfidf-cf-v1`), and Recommended Reels (`hybrid-tfidf-cf-v1`). Each model balances multi-objective signals including social graph topology, NLP term-frequency inverse-document-frequency (TF-IDF) cosine similarity, user-user collaborative filtering, creator relationship affinity, public engagement velocity, continuous exponential recency time decay, and greedy creator/hashtag diversity reranking.

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1 Introduction and Architectural Context

Insta-V2 employs a decoupled three-tier architecture separating client interaction, core business operations, and computationally intensive machine learning inference:

1. **Client Layer (React 19 + Vite):** Renders the responsive user interface, manages client-side media playback, tracks continuous reel viewing telemetry, and dispatches authenticated recommendation requests.
2. **Core API Layer (Node.js + Express):** Authenticates sessions via HTTP-only JWT cookies, pre-filters candidate sets from MongoDB Atlas (enforcing follow states, pending requests, and privacy boundaries), manages in-memory Redis/LRU candidate pools, and executes non-mutating Fisher-Yates freshness shuffling.
3. **Machine Learning Microservice Layer (Python + Django):** Houses the mathematical ranking engines implemented in `recommendations/engine.py`. The service communicates via internal HTTP POST endpoints guarded by shared-secret HMAC header verification (`X-Internal-Service-Key`).

1.1 Candidate Pool Caching and Freshness Shuffling

To ensure predictable sub-20ms API response times and protect CPU resources from redundant vector operations, the Node.js layer retrieves and caches candidate pools:

- **Users Pool:** 50 candidates (`RECOMMENDATION_POOL_SIZE_USERS = 50`)
- **Posts Pool:** 80 candidates (`RECOMMENDATION_POOL_SIZE_POSTS = 80`)
- **Reels Pool:** 60 candidates (`RECOMMENDATION_POOL_SIZE_REELS = 60`)

When a user refreshes their feed (`page=1` or `refresh=true`), Node.js applies an unbiased in-memory Fisher-Yates shuffle to shallow-cloned pools and advances session sequence pointers. The Python ML engine is invoked exclusively upon cache misses or explicit cache TTL expiration (300 seconds).

2 Global Notation and Normalization Principles

2.1 Universal Feature Normalization Operator

Before individual scoring signals are combined into composite rank scores, each signal vector $\mathbf{A} \in \mathbb{R}^N$ across candidate batch $\{1, \ldots, N\}$ is scaled to the closed unit interval $[0.0, 1.0]$ via the universal normalization operator $\text{Normalize}(\mathbf{A})$:

$$\text{Normalize}(\mathbf{A})_k = \begin{cases} \dfrac{A_k}{\max_j A_j}, & \text{if } \min_j A_j \geq 0 \ \wedge \ \max_j A_j > 10^{-9} \\ \dfrac{A_k - \min_j A_j}{\max_j A_j - \min_j A_j}, & \text{if } \min_j A_j < 0 \ \wedge \ (\max_j A_j - \min_j A_j) > 10^{-9} \\ 0.0, & \text{otherwise} \end{cases} \quad (1)$$

Mathematical Interpretation: For all candidate scoring functions where values are non-negative ($A_j \geq 0$), the transformation rescales each element relative to the highest-scoring candidate in the active batch. If negative values are present, min-max feature scaling is applied. If the entire batch evaluates to zero or is numerically degenerate ($< 10^{-9}$), a zero vector is returned. This guarantees that all sub-score distributions are scale-invariant prior to linear weighting.

3 Recommended Users Engine (`user-graph-interest-v1`)

Entry Function: `generate_user_recommendations(payload)`
Target Domain: Discovery of accounts for follow acquisition

3.1 Candidate Pre-Flight Exclusions

Let u denote the active authenticated user, and let $C = \{c_1, c_2, \ldots, c_N\}$ denote the candidate user accounts retrieved from MongoDB. The candidate pool strictly satisfies:

$$c \notin \{u\} \cup \text{Following}(u) \cup \text{Pending}(u) \cup \text{Blocked}(u) \tag{2}$$

where $\text{Following}(u)$ represents the accounts u already follows, $\text{Pending}(u)$ represents accounts with pending mutual follow requests, and $\text{Blocked}(u)$ represents bidirectional privacy blocks.

3.2 Scoring Components

3.2.1 Mutual Follower Graph Affinity (S_{mutual})

Let $m_c = |\,\text{Followers}(u) \cap \text{Followers}(c)|$ be the count of shared second-degree connections over the social graph. The topological affinity is defined as:

$$S_{\text{mutual}}(u, c) = \begin{cases} \min\left(1.0,\ 0.40 + 0.20 \cdot \log_2(1 + m_c)\right), & \text{if } m_c > 0 \\ 0.0, & \text{if } m_c = 0 \end{cases} \tag{3}$$

Interpretation: The initial existence of a single mutual follower ($m_c = 1$) awards an immediate base graph score of $0.40 + 0.20 \cdot \log_2(2) = 0.60$. Each subsequent doubling of mutual connections contributes an additional $+0.20$, saturating at 1.0 when $m_c \geq 7$.

3.2.2 Bio and Interest TF-IDF Cosine Similarity (S_{content})

Text documents are formed by concatenating biography text with normalized hashtag interest tokens:

$$D_u = \text{bio}_u \parallel \text{tags}_u, \qquad D_c = \text{bio}_c \parallel \text{tags}_c \tag{4}$$

Vectorization is performed using $\text{TfidfVectorizer}(\text{max_features} = 1000, \text{stop_words} = \text{"english"}, \text{ngram_range} = (1, 2))$ yielding feature vectors $\mathbf{v}_u, \mathbf{v}_c \in \mathbb{R}^K$. The unconstrained cosine similarity is:

$$\text{sim}_{\cos}(\mathbf{v}_c, \mathbf{v}_u) = \frac{\mathbf{v}_c \cdot \mathbf{v}_u}{\|\mathbf{v}_c\|_2 \|\mathbf{v}_u\|_2} \tag{5}$$

The bounded, normalized content similarity score is:

$$S_{\text{content}}(u, c) = \text{Normalize}\left(\text{clip}\left(\text{sim}_{\cos}(\mathbf{v}_c, \mathbf{v}_u),\ 0.0,\ 1.0\right)\right) \tag{6}$$

If either profile lacks textual metadata, $S_{\text{content}}(u, c) = 0.0$.

3.2.3 Prior Direct Interaction Affinity (S_{affinity})

Let $\text{Interactions}(u, c)$ be the sum of direct engagements (post likes, comments, and reel watch sessions) logged between user u and author c:

$$\begin{aligned} A_{\text{raw}}(u, c) &= \ln\left(1 + \text{Interactions}(u, c)\right) \\ S_{\text{affinity}}(u, c) &= \text{Normalize}\left(A_{\text{raw}}(u, c)\right) \end{aligned} \tag{7}$$

Logarithmic transformation $\ln(1 + x)$ dampens extreme interaction frequencies while penalizing zero-interaction accounts with an initial score of 0.0.

3.2.4 Account Popularity and Publishing Activity (S_{pop})

Evaluates platform reach and content publishing consistency:

$$\begin{aligned} P_{\text{raw}}(c) &= \ln\left(1 + \text{FollowersCount}(c) + 1.5 \cdot \text{PostsCount}(c)\right) \\ S_{\text{pop}}(c) &= \text{Normalize}\left(P_{\text{raw}}(c)\right) \end{aligned} \tag{8}$$

A coefficient of 1.5 is applied to post volume to boost active content creators over dormant accounts with static follower counts.

3.3 Composite User Scoring Equations

3.3.1 Warm User Formulation

For users who follow at least one account or possess logged interaction history:

$$S_{\text{user}}(u,c) = 0.35 \cdot S_{\text{mutual}}(u,c) + 0.25 \cdot S_{\text{content}}(u,c) + 0.20 \cdot S_{\text{affinity}}(u,c) + 0.20 \cdot S_{\text{pop}}(c) \tag{9}$$

Mutual graph connectivity carries the highest weight (35%), followed by topical similarity (25%), prior direct interactions (20%), and creator popularity (20%).

3.3.2 Cold-Start User Formulation

Triggered when $|\,\text{Following}(u)| = 0 \;\wedge\; |\,\text{InteractedAuthors}(u)| = 0$:

$$S_{\text{user}}(u,c) = 0.55 \cdot S_{\text{pop}}(c) + 0.35 \cdot S_{\text{content}}(u,c) + 0.10 \cdot S_{\text{affinity}}(u,c) \tag{10}$$

Because mutual graph connections do not yet exist ($S_{\text{mutual}} = 0$), global creator popularity (55%) and textual interest alignment (35%) drive initial exploration.

3.4 Recommendation Attribution Classification

Ranked candidates are annotated with human-readable attribution reasons evaluated by hierarchical threshold priority:

$$\text{Reason}(c) = \begin{cases} \text{"mutual_followers"}, & \text{if } m_c > 0 \\ \text{"interacted_recently"}, & \text{if } S_{\text{affinity}}(u,c) \geq 0.25 \\ \text{"shared_interests"}, & \text{if } S_{\text{content}}(u,c) \geq 0.20 \\ \text{"popular"}, & \text{if } S_{\text{pop}}(c) \geq 0.40 \;\vee\; \text{FollowersCount}(c) \geq 10 \\ \text{"suggested_for_you"}, & \text{otherwise} \end{cases} \tag{11}$$

4 Recommended Posts Engine (`hybrid-tfidf-cf-v1`, `posts`)

Entry Function: `generate_hybrid_recommendations(payload)`
Target Domain: Photo and carousel exploration feed

4.1 Candidate Pre-Flight Exclusions

Let $I = \{i_1, i_2, \ldots, i_N\}$ be the candidate posts pre-filtered from MongoDB, strictly excluding content from blocked creators and private accounts unless followed.

4.2 Historical Interaction Weighting

Engagements by active user u on historical items j are mapped to scalar interaction weights:

$$W(u,j) = 2.5 \cdot \mathbb{I}(j \in \text{Liked}(u)) + 3.5 \cdot \mathbb{I}(j \in \text{Commented}(u)) + 1.2 \cdot \mathbb{I}(\text{Author}(j) \in \text{FollowedAuthors}(u)) \tag{12}$$

where $\mathbb{I}(\cdot) \in \{0,1\}$ is the binary indicator function. Comments denote higher user commitment than likes and receive $+3.5$ versus $+2.5$. Items authored by creators the user follows receive $+1.2$.

4.3 Scoring Components

4.3.1 Content-Based TF-IDF Projection (S_{content})

Candidate documents $D_j = \text{caption}_j \,\|\, \text{tags}_j \,\|\, \text{loc}_j \,\|\, \text{media_type}_j$ are vectorized into a 1500-feature matrix $\mathbf{X} \in \mathbb{R}^{N \times M}$ using $\text{TfidfVectorizer}(\text{max_features} = 1500, \text{stop_words} = \text{"english"}, \text{ngram_range} = (1,2), \text{min_df} = 1)$.

The unnormalized user preference vector $\mathbf{u} \in \mathbb{R}^M$ and its unit Euclidean form $\mathbf{u}_{\text{norm}}$ are:

$$\mathbf{u} = \mathbf{X}^T \mathbf{w}_{\text{item}}, \qquad \mathbf{u}_{\text{norm}} = \frac{\mathbf{u}}{\|\mathbf{u}\|_2} \tag{13}$$

where $\mathbf{w}_{\text{item}} = [W(u,j)]_{j=1}^{N}$. Content similarity across all candidates is computed as:

$$S_{\text{content}}(u,i) = \text{Normalize}\left(\text{clip}\left(\mathbf{X}_i \cdot \mathbf{u}_{\text{norm}}^T,\ 0.0,\ 1.0\right)\right) \tag{14}$$

4.3.2 User-User Collaborative Filtering (S_{CF})

Let P denote the peer cohort sharing interaction history with u. Peer similarity over overlapping interaction sets $I_u \cap I_p$ is computed via:

$$\text{sim}_{\text{peer}}(u,p) = \frac{\sum_{j \in I_u \cap I_p} W(u,j) \cdot W(p,j)}{\|\mathbf{w}_u\|_2 \cdot \|\mathbf{w}_p\|_2} \tag{15}$$

Filtering peers below the significance threshold $\text{sim}_{\text{peer}}(u,p) > 0.05$, candidate score CF_i accumulates weighted peer interactions:

$$CF_i = \sum_{p:\text{sim}_{\text{peer}}(u,p)>0.05} \text{sim}_{\text{peer}}(u,p) \cdot W(p,i), \qquad S_{\text{CF}}(i) = \text{Normalize}(CF_i) \tag{16}$$

4.3.3 Creator Relationship Affinity (S_{creator})

Reflects direct author-level relationships:

$$S_{\text{creator}}(i) = \begin{cases} 1.0, & \text{if Author}(i) \in \text{FollowedAuthors}(u) \\ 0.5, & \text{if Author}(i) \in \text{InteractedAuthors}(u) \setminus \text{FollowedAuthors}(u) \\ 0.0, & \text{otherwise} \end{cases} \tag{17}$$

4.3.4 Public Engagement Popularity (S_{pop})

$$\begin{aligned} P_{\text{raw}}(i) &= \ln(1 + \text{Likes}(i) + 2.0 \cdot \text{Comments}(i) + 2.0 \cdot \text{Shares}(i)) \\ S_{\text{pop}}(i) &= \text{Normalize}(P_{\text{raw}}(i)) \end{aligned} \tag{18}$$

4.3.5 Exponential Recency Time Decay (S_{recency}, 72-Hour Half-Life)

Let $\Delta t_{\text{hours}} = \max\left(0.0,\ \frac{t_{\text{now}} - t_{\text{created}}}{3600}\right)$ be the item age in decimal hours:

$$\begin{aligned} R_{\text{raw}}(i) &= \exp\left(-\frac{\Delta t_{\text{hours}}}{72.0}\right) \\ S_{\text{recency}}(i) &= \text{Normalize}(R_{\text{raw}}(i)) \end{aligned} \tag{19}$$

4.4 Base Composite Scoring Equations

4.4.1 Warm User Base Score Formulation

For users with $|\mathbf{w}_{\text{item}}| \geq 2 \vee |\,\text{FollowedAuthors}(u)| > 0$:

$$S_{\text{base}}(i) = 0.32 \cdot S_{\text{content}}(i) + 0.25 \cdot S_{\text{CF}}(i) + 0.20 \cdot S_{\text{creator}}(i) + 0.13 \cdot S_{\text{pop}}(i) + 0.10 \cdot S_{\text{recency}}(i) \tag{20}$$

4.4.2 Cold-Start Base Score Formulation

Triggered when $|\mathbf{w}_{\text{item}}| < 2 \wedge |\,\text{FollowedAuthors}(u)| = 0$:

$$S_{\text{base}}(i) = 0.45 \cdot S_{\text{pop}}(i) + 0.45 \cdot S_{\text{recency}}(i) + 0.10 \cdot S_{\text{content}}(i) \tag{21}$$

4.5 Already-Interacted Item Demotion

To promote novelty and eliminate feed repetition, previously engaged items incur a 60% score reduction:

$$S_{\text{base}}(i) \leftarrow S_{\text{base}}(i) \cdot 0.40 \qquad \text{if } i \in (\text{Liked}(u) \cup \text{Commented}(u)) \tag{22}$$

4.6 Iterative Creator and Hashtag Diversity Reranking

A naive ranking by S_{base} frequently clusters multiple posts from a single prolific author. The reranker operates greedily in an iterative sequence:

$$S_{\text{adjusted}}(i) = (S_{\text{base}}(i) + 0.15) \cdot 0.55^{C(\text{Author}(i))} \cdot \frac{1}{1.0 + 0.15 \cdot \sum_{t \in \text{tags}_i[:3]} H(t)} \tag{23}$$

where:

- $C(\text{Author}(i))$ is the count of items from $\text{Author}(i)$ already selected in the active session.
- $H(t)$ is the selection frequency of hashtag t in the top-3 hashtag positions across selected items.
- $+0.15$ is a baseline additive offset preserving rank diversity for lower-scoring candidates.

5 Recommended Reels Engine (`hybrid-tfidf-cf-v1`, `reels`)

Entry Function: `generate_hybrid_recommendations(payload)`
Target Domain: Vertical short-form video feed

5.1 Video Telemetry and Implicit Interaction Weighting

For candidate reel r, client beacons transmit playback duration and total media duration:

$$\text{CompletionRate}(u, r) = \min\left(1.0,\ \frac{\text{WatchDurationSeconds}(u, r)}{\text{VideoDurationSeconds}(r)}\right) \tag{24}$$

Telemetry qualifies as an intentional interaction if either threshold is reached:

$$\text{Qualifies}(u, r) \iff \text{WatchDurationSeconds}(u, r) \geq 2.0 \ \vee \ \text{CompletionRate}(u, r) \geq 0.25 \quad (25)$$

Qualified view events generate an implicit watch weight:

$$W_{\text{watch}}(u, r) = 0.5 + 2.5 \cdot \text{CompletionRate}(u, r) \quad (26)$$

This scalar weight (0.5 to 3.0) accumulates into $\mathbf{w}_{\text{item}}$ alongside explicit likes (+2.5) and comments (+3.5).

5.2 Acoustic and Audio Track Affinity

Video background music and ambient sounds are tokenized into boosted feature tokens:

$$\text{audio_tokens} = \text{"audio_"} + \text{audio_track.replace(' ', '_')} + \text{" "} + \text{audio_track} \quad (27)$$

These tokens are injected into candidate document D_r, enabling content cosine similarity $S_{\text{content}}(u, r)$ to cluster reels utilizing identical audio trends.

5.3 Accelerated Recency Decay (48-Hour Half-Life)

Short-form video trends evolve rapidly. Reels enforce an accelerated 48-hour half-life ($T_{1/2} = 48.0$ hours):

$$\begin{aligned} R_{\text{raw}}(r) &= \exp\left(-\frac{\Delta t_{\text{hours}}}{48.0}\right) \\ S_{\text{recency}}(r) &= \text{Normalize}(R_{\text{raw}}(r)) \end{aligned} \quad (28)$$

5.3.1 Explicit Half-Life Comparison

- **Posts Half-Life ($T_{1/2} = 72.0$ hours):** $R(72\text{h}) = 0.50,\ R(144\text{h}) = 0.25$.
- **Reels Half-Life ($T_{1/2} = 48.0$ hours):** $R(48\text{h}) = 0.50,\ R(96\text{h}) = 0.25$.

5.4 Composite Scoring and Diversity Damping

Reels utilize identical composite weights ($0.32S_{\text{content}} + 0.25S_{\text{CF}} + 0.20S_{\text{creator}} + 0.13S_{\text{pop}} + 0.10S_{\text{recency}}$), $0.40\times$ interaction demotion, and iterative creator/hashtag diversity damping ($0.55^C \cdot \frac{1}{1+0.15\sum H(t)}$) as the Posts pipeline.

6 Mathematical Notation Reference Table

7 Implementation Summary and Verification

7.1 Code Source Verification

All equations documented in this specification were verified against the production code in `ml-service/recommendations/engine.py`:

- Normalization method: Evaluates batch maximum scaling relative to 10^{-9} threshold (§1).
- Mutual follower formula: Validated with logarithmic base 2 and $0.40 + 0.20 \cdot \log_2(1 + m_c)$ progression (§3).
- Interaction weights: Validated at +2.5 for likes, +3.5 for comments, +1.2 for followed authors, and $0.5 + 2.5 \cdot \text{CompletionRate}$ for reel watches (§12, §26).

Table 1: Summary of Mathematical Symbols and Implementation Bindings

Symbol	Domain	Implementation Definition in `engine.py`
u	String	Active authenticated user identifier.
c	String	Candidate user account identifier under evaluation.
p	String	Peer user in collaborative filtering cohort sharing interactions with u.
i	String	Candidate post item identifier.
r	String	Candidate reel video item identifier.
j	String	Arbitrary item index in historical interaction sets ($j \in I_u \cup I_p$).
m_c	$\mathbb{Z}_{\geq 0}$	Mutual follower count between active user u and candidate c.
$S_{\mathbf{mutual}}$	$[0.0, 1.0]$	Normalized second-degree graph affinity score based on m_c.
$S_{\mathbf{content}}$	$[0.0, 1.0]$	Normalized cosine similarity between user profile and candidate item.
$S_{\mathbf{affinity}}$	$[0.0, 1.0]$	Normalized direct interaction affinity score between user u and candidate c.
$S_{\mathbf{pop}}$	$[0.0, 1.0]$	Normalized logarithmic public engagement velocity score.
$S_{\mathbf{CF}}$	$[0.0, 1.0]$	Normalized user-user collaborative filtering score from peer cohort.
$S_{\mathbf{creator}}$	$\{0.0, 0.5, 1.0\}$	Discrete affinity for followed (1.0), interacted (0.5), or unlinked (0.0) creator.
$S_{\mathbf{recency}}$	$[0.0, 1.0]$	Normalized exponential time decay score based on item age and half-life.
$W(u, j)$	$\mathbb{R}_{\geq 0}$	Cumulative scalar interaction weight assigned by user u to item j.
I_u, I_p	Set of IDs	Sets of item IDs with recorded engagements by user u and peer p.
$\mathbf{X}$	$\mathbb{R}^{N \times M}$	1500-column TF-IDF candidate feature matrix.
$\mathbf{w}_{\mathbf{item}}$	$\mathbb{R}^N$	Dense vector of interaction weights used to project user profile $\mathbf{u} = \mathbf{X}^T \mathbf{w}_{\text{item}}$.
$C(\text{Author})$	$\mathbb{Z}_{\geq 0}$	Count of items from an author already chosen in active session.
$H(t)$	$\mathbb{Z}_{\geq 0}$	Cumulative frequency of hashtag t across chosen items in top-3 slots.
$\Delta t_{\mathbf{hours}}$	$\mathbb{R}_{\geq 0}$	Elapsed age of candidate item in decimal hours since publication.

- Collaborative filtering: Validated with peer similarity cutoff at 0.05 (§16).
- Recency decay: Validated at $T_{1/2} = 72.0$ hours for posts and $T_{1/2} = 48.0$ hours for reels (§19, §28).
- Diversity reranking: Validated with 0.55^C author penalty and $\frac{1}{1+0.15\sum H(t)}$ tag damping (§23).

7.2 Automated Integration Test Suite

The integrity of the three pipelines is continuously validated via the automated backend test suite:

```
npm run test:recommendations --prefix server
```

The suite validates candidate generation, cache storage, TTL expiration, per-user isolation, Fisher-Yates freshness shuffling, and fault-tolerant fallback modes under live microservice conditions.